The Pre-Lecture Analysis Sheet

What It Is · Why It Works · How to Use It

> **"Why should I fill this out if I'm just going to learn it in class?"**
>
> — Every student who ever got a lower grade than expected.

Fair question. The answer is deceptively simple: **your brain learns better the second time it sees something than the first.** The Pre-Lecture Analysis Sheet (PLAS) makes sure the lecture is never truly the first time. By the time you walk in, you've already wrestled with the material — activated what you know, flagged what you don't, and primed your brain to absorb what the lecture fills in. This isn't busywork. It's the oldest trick in the cognitive science playbook.

The Research: This Actually Works

The PLAS is grounded in **interteaching** — a behavioral approach to learning pioneered by **Saville, Zinn, and Elliott (2005)** and extended in multiple controlled classroom studies. Preparation guides were identified as a core driver of performance gains, with students who completed them before class consistently outscoring lecture-only peers.

+10	76%	3×
percentage-point gain on exams vs. lecture-only (Saville et al., 2006)	correct on final exam with prep guides vs. 69.5% lecture-only (Scoboria & Pascual-Leone, 2009)	exposures to material: before, during, after — the spacing effect at work

Critically, Saville (2006) found that **prep guides alone are not enough** — the gains come when preparation is paired with active class engagement. The PLAS is built for exactly that: sections to complete before, during, and after so all three exposures happen.

The Six Sections — What Each One Does

A Before Learning Targets — I Can Statements

Rate your confidence 1–5 honestly. Low scores aren't failure — they're targeting information. Re-rate after lecture. The Pre/Post gap shows you exactly where you grew and where you still need work.

B Before Activate — Connect to What You Know

Prior knowledge is the scaffold new knowledge hangs on. B1 links today to last week; B2 makes those links explicit. Even imperfect retrieval strengthens memory traces — guessing is not wasted.

C Before Vocabulary Pre-Load

Write your own definition. It doesn't have to be right — it just has to be yours. When the lecture corrects it, that moment of correction is encoded far more deeply than a passive copy ever would be.

D During Anticipatory Questions

Leave blank before class. Fill in during. These questions match exactly what the lecture will cover. Instead of transcribing slides, you're answering targeted questions in real time — a dramatically stronger encoding strategy.

E Before + After Clinical Case

Attempt the Pre questions with your current knowledge. Return to Post questions after lecture. The contrast between your two attempts makes learning visible to yourself — you can literally see what the class taught you.

F After Synthesis & Self-Monitoring

Complete within 24 hours. The three-column table (Confident / Unclear / Connected) is a metacognitive tool — you're not just reviewing content, you're monitoring your own understanding. Students who self-monitor consistently outperform those who don't (Dunning & Kruger's useful flip side).

> **The most common mistake:** completing Section D before class by looking up answers. This feels productive but defeats the purpose — it turns active learning into passive reading and eliminates the real-time encoding benefit. Leave Section D blank. Trust the process.

> **One sheet. Any class.** The PLAS format is consistent across every course, so the method becomes automatic. You're not learning a new system each semester — you're applying the same high-performance study habit to different content. Build the habit in one class; it transfers to all of them.

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